

A BAYESIAN ANALYSIS OF SOME NONPARAMETRIC PROBLEMS¹

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1. Introduction and summary. The Bayesian approach to statistical problems, though fruitful in many ways, has been rather unsuccessful in treating nonparametric problems. This is due primarily to the difficulty in finding workable prior distributions on the parameter space, which in nonparametric problems is taken to be a set of probability distributions on the sample space. There are two desirable properties of a prior distribution for nonparametric problems.

(I) The support of the prior distribution should be compact with respect to some suitable topology on the space of probability distributions.

(II) Posterior distributions should be proper and the prior probability distribution should be noninformative.



MIXTURES OF DIRICHLET PROCESSES WITH APPLICATIONS TO BAYESIAN NONPARAMETRIC STATISTICS

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A random process called the Dirichlet process whose sample functions are almost surely probability measures has been proposed as an approach to analyzing nonparametric problems from a Bayesian point of view.

An important result obtained by Ferguson in this approach is that if observations are made on a random variable whose distribution is a random sample function of a Dirichlet process, then the conditional distribution

The partition from a DP is just a re-write of the Ewens formula!

$$\mathbb{P}(n_1, \dots, n_y) = \frac{\alpha^y}{(\alpha)_n} \prod_{j=1}^y (n_j - 1)!$$

Chinese Restaurant Process
(Blackwell and MacQueen, 1973)

$$X_{n+1} = \begin{cases} X_j^* & \text{w. p. } n_j/(\alpha + n) \\ \text{new} & \text{w. p. } \alpha/(\alpha + n) \end{cases}$$

Ferguson
(1973)
Antoniak
(1974)

Fisher, Corbet
and Williams
(1943)

Good
(1953)

Good and
Toulmin
(1956)

Ewens
(1972)

Dirichlet
process

Fisher's log-
series model

Prediction of N.
of species in a
further sample

Ewens'
sampling
formula